(b)	**Inheritance**
Students should:	
3.14	understand that the genome is the entire DNA of an organism and that a gene is a section of a molecule of DNA that codes for a specific protein
3.15	understand that the nucleus of a cell contains chromosomes on which genes are located
3.16B	**describe a DNA molecule as two strands coiled to form a double helix, the strands being linked by a series of paired bases: adenine (A) with thymine (T), and cytosine (C) with guanine (G)**
3.17B	**understand that an RNA molecule is single stranded and contains uracil (U) instead of thymine (T)**
3.18B	**describe the stages of protein synthesis including transcription and translation, including the role of mRNA, ribosomes, tRNA, codons and anticodons**
3.19	understand how genes exist in alternative forms called alleles which give rise to differences in inherited characteristics
3.20	understand the meaning of the terms: dominant, recessive, homozygous, heterozygous, phenotype, and genotype
3.21B	**understand the meaning of the term codominance**
3.22	understand that most phenotypic features are the result of polygenic inheritance rather than single genes
3.23	describe patterns of monohybrid inheritance using a genetic diagram
3.24	understand how to interpret family pedigrees
3.25	predict probabilities of outcomes from monohybrid crosses
3.26	understand how the sex of a person is controlled by one pair of chromosomes, XX in a female and XY in a male
3.27	describe the determination of the sex of offspring at fertilisation, using a genetic diagram
3.28	understand how division of a diploid cell by mitosis produces two cells that contain identical sets of chromosomes
3.29	understand that mitosis occurs during growth, repair, cloning and asexual reproduction
3.30	understand how division of a cell by meiosis produces four cells, each with half the number of chromosomes, and that this results in the formation of genetically different haploid gametes
3.31	understand how random fertilisation produces genetic variation of offspring
3.32	know that in human cells the diploid number of chromosomes is 46 and the haploid number is 23
3.33	understand that variation within a species can be genetic, environmental, or a combination of both
3.34	understand that mutation is a rare, random change in genetic material that can be inherited

Students should:	
3.35B	**understand how a change in DNA can affect the phenotype by altering the sequence of amino acids in a protein**
3.36B	**understand how most genetic mutations have no effect on the phenotype, some have a small effect and rarely do they have a significant effect**
3.37B	**understand that the incidence of mutations can be increased by exposure to ionising radiation (for example, gamma rays, x-rays and ultraviolet rays) and some chemical mutagens (for example, chemicals in tobacco)**
3.38	explain Darwin's theory of evolution by natural selection
3.39	understand how resistance to antibiotics can increase in bacterial populations, and appreciate how such an increase can lead to infections being difficult to control